



LOVELY PROFESSIONAL UNIVERSITY

School of Computer Science and Engineering

SUMMER TRAINING/INTERNSHIP

PROJECT REPORT

(Term June–July 2025)

Demand Forecasting System for Bike-Sharing Services

Submitted by

NAME	REGISTER NUMBER
Yadhunaath B	12401067
Deepanshu Yadav	12405819
Ridhima	12407097

Course Code : PETV149

Under the Guidance of

Miss.Savleen Kaur

MissSandeep Kaur

School of Computer Science and Engineering

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Project Team:

- Yadhunaath B (12401067)
- Deepanshu Yadav (12405819)
- Ridhima (12407097)

Table of Contents

Chapter 1: Introduction

- 1.1 Project Profile
- 1.2 Overview of Training Domain
- 1.3 Objective of the Project

Chapter 2: Training Overview

- 2.1 Tools & Technologies Used
- 2.2 Areas Covered During Training
- 2.3 Summary of Work

Chapter 3: Project Details

- 3.1 Title of the Project
- 3.2 Problem Definition
- 3.3 Scope and Objectives
- 3.4 System Requirements
- 3.5 Data Source Description

Chapter 4: Implementation

- 4.1 Tools and Technologies
- 4.2 Methodology
- 4.3 Modules and Implementation Details
 - 4.3.1 Data Preprocessing and Feature Engineering
 - 4.3.2 Exploratory Data Analysis (EDA)
 - 4.3.3 Machine Learning Model Development and Evaluation
 - 4.3.4 Power BI Dashboard Development

Chapter 5: Results and Discussion

- 5.1 Output and Results
 - 5.1.1 Key Insights from EDA
 - 5.1.2 Machine Learning Model Performance
 - 5.1.3 Power BI Dashboard Overview
- 5.2 Challenges Faced
- 5.3 Key Learnings
- 5.4 Entire Code Snippet

Chapter 6: Conclusion and Future Scope

- 6.1 Summary
- 6.2 Conclusion
- 6.3 Future Scope

Bibliography & References

Chapter 1: Introduction

1.1 Project Profile

This project was undertaken as part of the **June–July Summer Training 2026** conducted by the School of Computer Science and Engineering, Lovely Professional University.

The Summer Training program is designed to provide students with practical exposure to real-world problem-solving by applying the concepts learned during their academic coursework. Throughout the training period, students work on industry-oriented projects using modern tools and technologies, helping them strengthen their technical knowledge, analytical thinking, and implementation skills.

As part of the Summer Training conducted from **June 2026 to July 2026**, this project focuses on developing a **Demand Forecasting System for Bike-Sharing Services**. The work involved data preprocessing, exploratory data analysis, machine learning model development, performance evaluation, and interactive dashboard creation using **Python** and **Microsoft Power BI**. The project provided hands-on experience in data analytics, machine learning, and business intelligence while enhancing practical problem-solving and project development skills.

1.2 Overview of Training Domain

. The project was carried out in the domain of **Data Analytics and Business Intelligence**, with a focus on analyzing historical bike-sharing data to identify usage patterns and support data-driven decision-making. The training involved data cleaning, preprocessing, exploratory data analysis (EDA), visualization, and dashboard development using **Python** and **Microsoft Power BI**. By analyzing historical rental trends, weather conditions, time-based variables, and user demographics, the project aimed to generate meaningful insights for demand analysis and operational planning.

1.3 Objective of the Project

. The primary objective of this project was to **analyze historical bike-sharing data** and develop an interactive **Bike Sharing Demand Analysis Dashboard** using **Power BI**. The project involved cleaning and preprocessing the dataset, performing Exploratory Data Analysis (EDA), identifying the impact of weather conditions, time-based variables, and user demographics on bike rentals, and visualizing key performance indicators through interactive dashboards. The dashboard provides demand forecasts, peak-hour analytics, usage trends, and business insights to support inventory management and scheduling decisions.

Chapter 2: Training Overview

2.1 Tools & Technologies Used

- Programming & ML: Python, Pandas, NumPy, Scikit-learn
- Data Visualization: Matplotlib, Seaborn, Microsoft Power BI
- Development Environment: Jupyter Notebook / VS Code

2.2 Areas Covered During Training

- Data collection and dataset understanding
- Data cleaning and preprocessing
- Feature engineering using date and time variables
- Exploratory Data Analysis (EDA)
- Correlation analysis and visualization
- Machine Learning using Linear Regression
- Model evaluation using MAE, RMSE, and R^2 Score
- Interactive dashboard development using Microsoft Power BI

2.3 Summary of Work

The project followed a systematic workflow to analyze and predict bike rental demand using historical data.

The work began with importing the **Seoul Bike Sharing Demand** dataset into Python. The dataset was inspected for missing values and duplicate records. The **Date** column was converted into datetime format, duplicate rows were removed, and records containing invalid dates were discarded. For better readability and easier processing, several columns such as **Temperature**, **Humidity**, **WindSpeed**, **DewPoint**, **SolarRadiation**, **Rainfall**, **Snowfall**, **FunctioningDay**, and **BikeCount** were renamed.

After preprocessing, feature engineering was performed by extracting useful time-based variables from the Date column. New features including **Month**, **DayOfWeek**, **Weekend**, and **PeakHour** were created to improve data analysis and prediction. Exploratory Data Analysis (EDA) was then carried out to understand the characteristics of the dataset. Various visualizations such as histograms, correlation heatmaps, scatter plots, boxplots, and hourly trend graphs were created to study the influence of weather conditions, seasons, and time on bike rental demand. For demand prediction, a **Linear Regression** model was developed using a Scikit-learn **Pipeline**. A **ColumnTransformer** was used to apply One-Hot Encoding to categorical variables while numerical features were passed directly to the model. The dataset was divided into **80% training data** and **20% testing data** using chronological splitting to preserve the time sequence. The performance of the model was evaluated using **Mean Absolute Error (MAE)**, **Root Mean Squared Error (RMSE)**, and **R^2 Score**. Additional visualizations including **Actual vs Predicted**, **Residual Plot**, and **Residual Distribution** were generated to assess the prediction quality. Finally, the processed data and analytical insights were used to develop an interactive **Power BI dashboard**, enabling users to explore rental trends, seasonal demand, weather influence, and hourly usage patterns through interactive visualizations.

Chapter 3: Project Details

3.1 Title of the Project

Demand Forecasting System for Bike-Sharing Services

3.2 Problem Definition

Bike-sharing operators face persistent challenges in matching bike supply with fluctuating rider demand across hours, days, and seasons. Under-provisioning during peak periods leads to lost rides and poor user experience, while over-provisioning increases idle fleet, rebalancing, and maintenance costs. This project addresses the need for a data-driven demand forecasting system that predicts hourly rental volume from weather, time, and calendar variables, enabling operators to make informed inventory and staff-scheduling decisions.

3.3 Scope and Objectives

Scope: The project covers the analysis of a historical, hourly bike-rental dataset, the development and comparison of multiple regression models for demand prediction, and the creation of a Power BI dashboard prototype for forecast and peak-hour visualisation.

Objectives:

- To preprocess historical rental, weather, and calendar data.
- To perform EDA to identify key time-based and weather-based factors influencing bike demand.
- To engineer relevant features such as peak-hour flags, weekend indicators, and seasonal groupings.
- To build, train, and evaluate multiple regression models for hourly demand prediction.
- To design a Power BI dashboard to visualise demand forecasts, peak-hour analytics, and usage influencers.

3.4 System Requirements

The system must process a structured CSV dataset of historical rental records, clean and engineer features from it, build a regression model with acceptable predictive accuracy, and provide an interactive dashboard with filters and key demand metrics.

3.5 Data Source Description

The dataset used is the Seoul Bike Sharing Demand dataset, containing 8,760 hourly records across 14 original columns, covering one full year of rental counts alongside weather measurements (temperature, humidity, wind speed, visibility, dew point, solar radiation, rainfall, snowfall) and operational/calendar flags (season, holiday status, functioning day).

Dataset Name	Seoul Bike Sharing Demand Dataset
Total Records	8,760
Target Variable	Rented Bike Count
Data Type	Structural Tabular Form

File Format	CSV
--------------------	------------

Chapter 4: Implementation

4.1 Tools and Technologies

As listed in Chapter 2.1.

4.2 Methodology

Dataset Collection → Data Cleaning → Feature Engineering → Exploratory Data Analysis → Machine Learning Model Development → Model Evaluation → Power BI Dashboard Development

4.3 Modules and Implementation Details

4.3.1 Data Preprocessing and Feature Engineering

- The first stage of implementation involved preparing the dataset for analysis and prediction. The Seoul Bike Sharing Demand dataset was imported into Python using the Pandas library. Initial exploration was performed to understand the structure of the dataset by displaying the first few records, descriptive statistics, missing values, and duplicate records.
- The **Date** column was converted into the datetime format to enable extraction of useful temporal information. Duplicate records were removed from the dataset, and any records containing invalid dates were discarded to maintain data quality.
- To improve readability, several columns were renamed. For example: □ Temperature(°C) → Temperature ,Humidity(%) → Humidity

After preprocessing, four new features were created from the Date and Hour columns:

- **Month** – Name of the month extracted from the Date column.
- **DayOfWeek** – Day name extracted from the Date column.
- **Weekend** – Binary feature indicating whether the day is Saturday or Sunday.
- **PeakHour** – Binary feature indicating peak commuting hours (7–9 AM and 5–8 PM).
-

4.3.2 Exploratory Data Analysis (EDA)

1. Distribution of Bike Rentals

A histogram with a Kernel Density Estimate (KDE) curve was plotted to examine the distribution of hourly bike rental counts. This helped understand the frequency of different rental volumes.

2. Correlation Heatmap

A correlation heatmap was generated using all numerical variables. The heatmap illustrates the strength and direction of relationships among numerical features and the target variable (BikeCount), helping identify influential predictors.

3. Temperature vs Bike Rentals

A scatter plot was created to analyze the relationship between temperature and bike rental demand. The visualization shows how rental activity changes with temperature.

4. Humidity vs Bike Rentals

Another scatter plot was generated to examine the relationship between humidity and bike rental demand.

5. Bike Rentals by Season

A boxplot was used to compare bike rental distributions across the four seasons:

Spring
Summer
Autumn
Winter

6. Average Bike Rentals by Hour

The average number of rented bikes for each hour of the day was calculated and displayed using a line chart. This visualization identifies peak rental hours and daily demand patterns.

4.3.3 Machine Learning Model Development and Evaluation

After completing data preprocessing and exploratory analysis, a machine learning model was developed to predict hourly bike rental demand.

- **Feature Selection**

The following variables were selected as predictor features:

Hour ,Temperature ,Humidity ,WindSpeed ,Visibility ,DewPoint ,SolarRadiation ,Rainfall ,Snowfall ,Seasons ,Holiday ,FunctioningDay DayOfWeek ,Weekend ,PeakHour. The target variable for prediction was **BikeCount**.

- **Data Preprocessing for Machine Learning**

The predictor variables were divided into two categories:

Categorical Features

Seasons
Holiday
FunctioningDay
DayOfWeek

These variables were transformed using **One-Hot Encoding**.

- **Data Splitting**

The dataset was divided chronologically into:

80% Training Data

20% Testing Data

Chronological splitting was chosen to preserve the natural time order of the observations.

- **Model Development**

A **Scikit-learn Pipeline** was created consisting of two stages:

Data preprocessing using the ColumnTransformer.

Linear Regression model.

The pipeline was trained using the training dataset and then used to predict bike rental demand on the testing dataset.

- **Model Evaluation**

The prediction accuracy of the Linear Regression model was evaluated using three regression metrics:

Mean Absolute Error (MAE) – Measures the average prediction error.

Root Mean Squared Error (RMSE) – Measures prediction error while penalizing larger errors.

R² Score – Measures how well the model explains the variation in bike rental demand.

- To further evaluate the model, three visualizations were generated:
Actual vs Predicted Plot
Residual Plot
Residual Distribution
Finally, a new dataset containing the **Actual** and **Predicted** bike rental counts was exported as **bike_predictions.csv** for further analysis and visualization.

4.3.4 Power BI Dashboard Development

- After completing the machine learning implementation, the processed dataset and prediction results were imported into Microsoft Power BI to create an interactive dashboard.
The dashboard presents important information related to bike rental demand through various charts and KPIs. Interactive filters allow users to analyze rental patterns based on different factors such as season and hour.
- The Power BI dashboard includes visualizations for:
Overall bike rental demand
Seasonal rental patterns
Hourly rental trends
Weather-related analysis
Predicted versus actual bike rentals
Key performance indicators (KPIs)
- The dashboard enables users to explore the data interactively and supports informed decision-making for bike-sharing operations.

Chapter 5: Results and Discussion

5.1 Output and Results

The developed Demand Forecasting System was successfully implemented using Python and Microsoft Power BI. After preprocessing the dataset and performing feature engineering, Exploratory Data Analysis (EDA) was carried out to understand the factors affecting bike rental demand. A Linear Regression model was then trained to predict hourly bike rentals. The performance of the model was evaluated using standard regression metrics, and the processed data was visualized through an interactive Power BI dashboard.

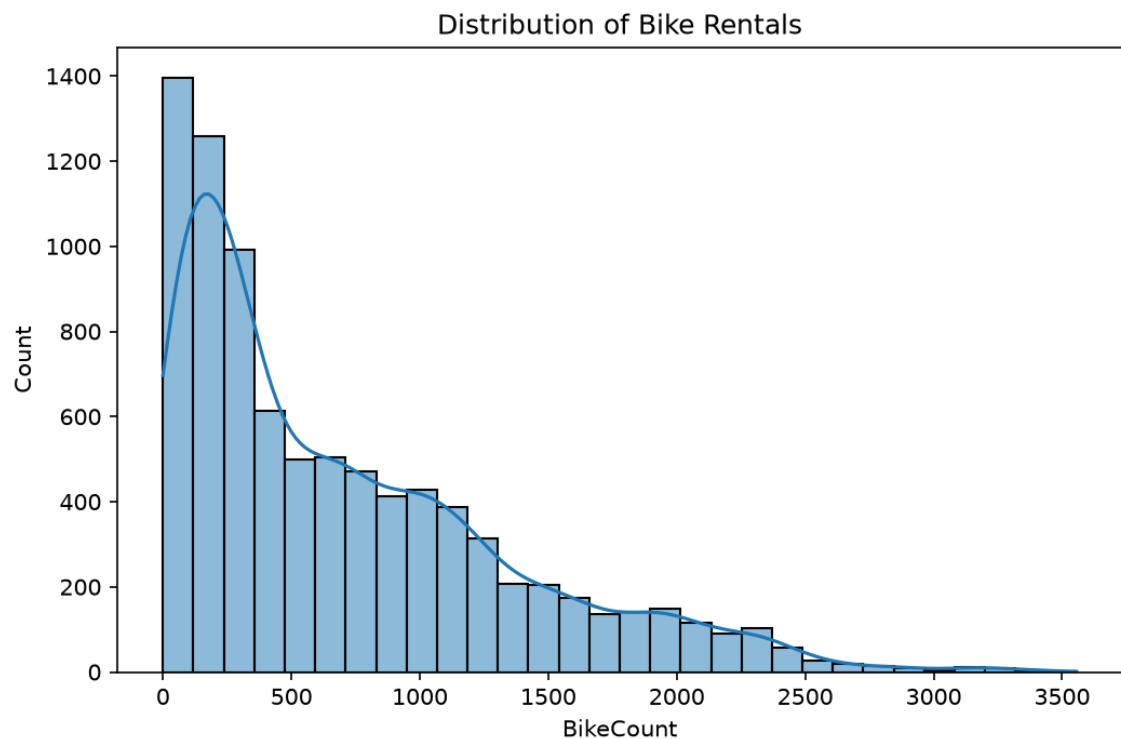
5.1.1 Key Insights from EDA

Exploratory Data Analysis (EDA) was performed to understand the characteristics of the dataset and identify the major factors influencing bike rental demand. Various visualizations such as histograms, correlation heatmaps, scatter plots, boxplots, and line charts were generated using Matplotlib and Seaborn.

1. Distribution of Bike Rentals

A histogram with a Kernel Density Estimate (KDE) curve was plotted to examine the distribution of hourly bike rental counts. **Observations:**

- The distribution is positively skewed, indicating that most hours record low to moderate bike rentals.
- A smaller number of observations show very high rental demand.
- The distribution confirms that bike demand varies significantly throughout the year.



2. Correlation Heatmap

Observations

- Temperature shows a positive relationship with bike rental demand.
- Hour also has a positive correlation with bike rentals, indicating higher demand during specific times of the day.
- Humidity has a negative relationship with bike demand.
- Rainfall and snowfall also show negative correlations, suggesting that adverse weather conditions reduce bike usage.
- The heatmap highlights weather conditions and time-based variables as important factors affecting bike rental demand.

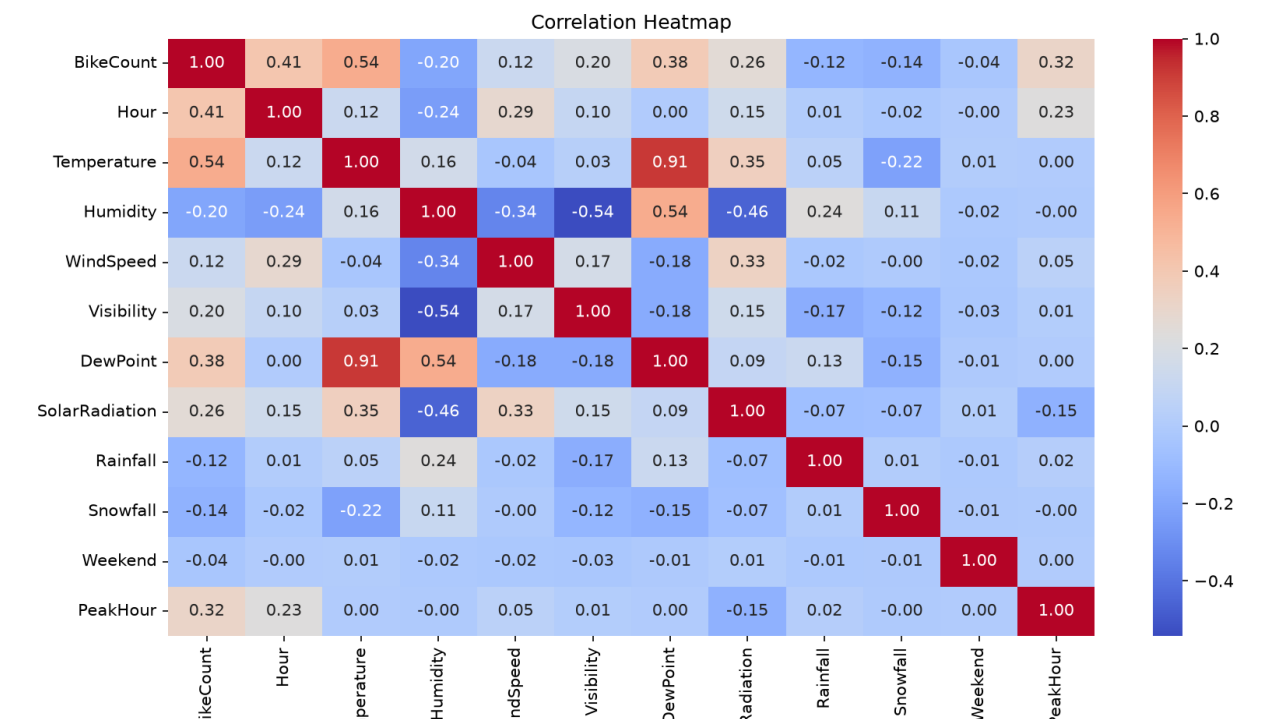


Figure 2: Correlation Heatmap

Correlation of Features with Bike Count

Feature	Correlation with BikeCount
Temperature	0.54
Hour	0.41
DewPoint	0.38
PeakHour	0.32

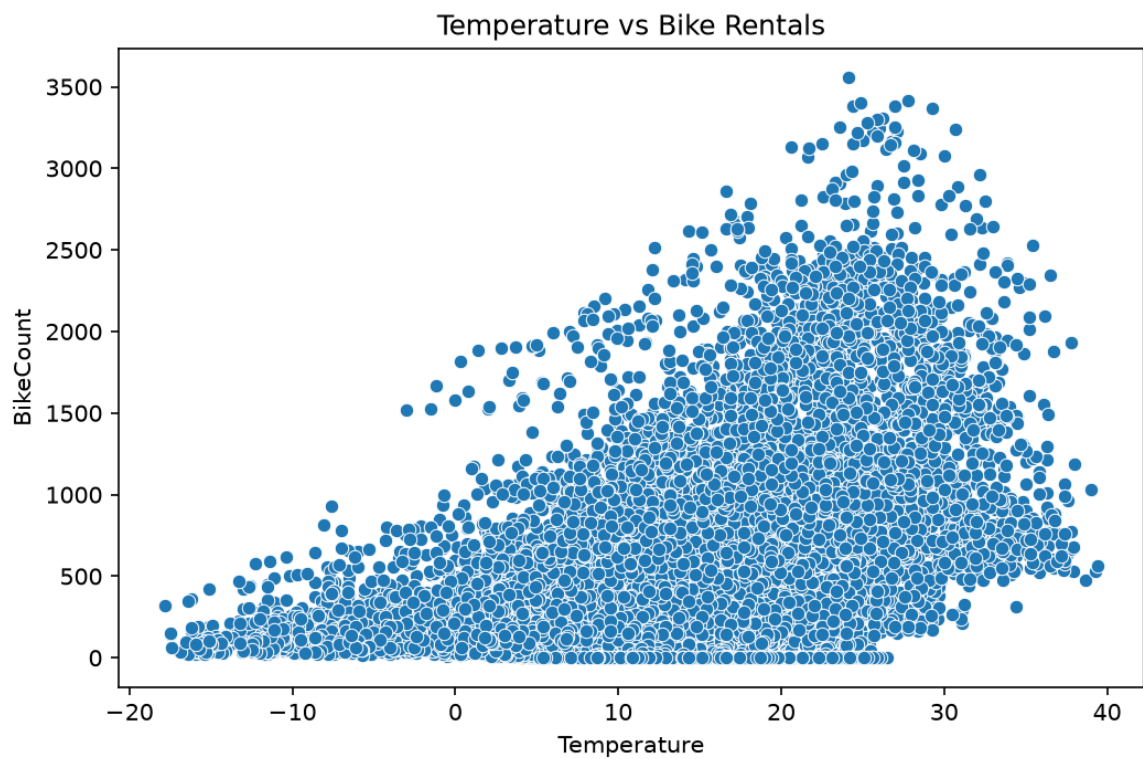
Feature	Correlation with BikeCount
SolarRadiation	0.26
Humidity	-0.2
Visibility	0.2
Snowfall	-0.14
Rainfall	-0.12
WindSpeed	0.12
Weekend	-0.04

3. Temperature vs Bike Rentals

A scatter plot was created to analyze the relationship between temperature and bike rental demand.

Observations

- Bike rentals generally increase as temperature rises.
- Warmer weather encourages greater use of bike-sharing services.
- Although some variation exists, an overall positive trend is visible.

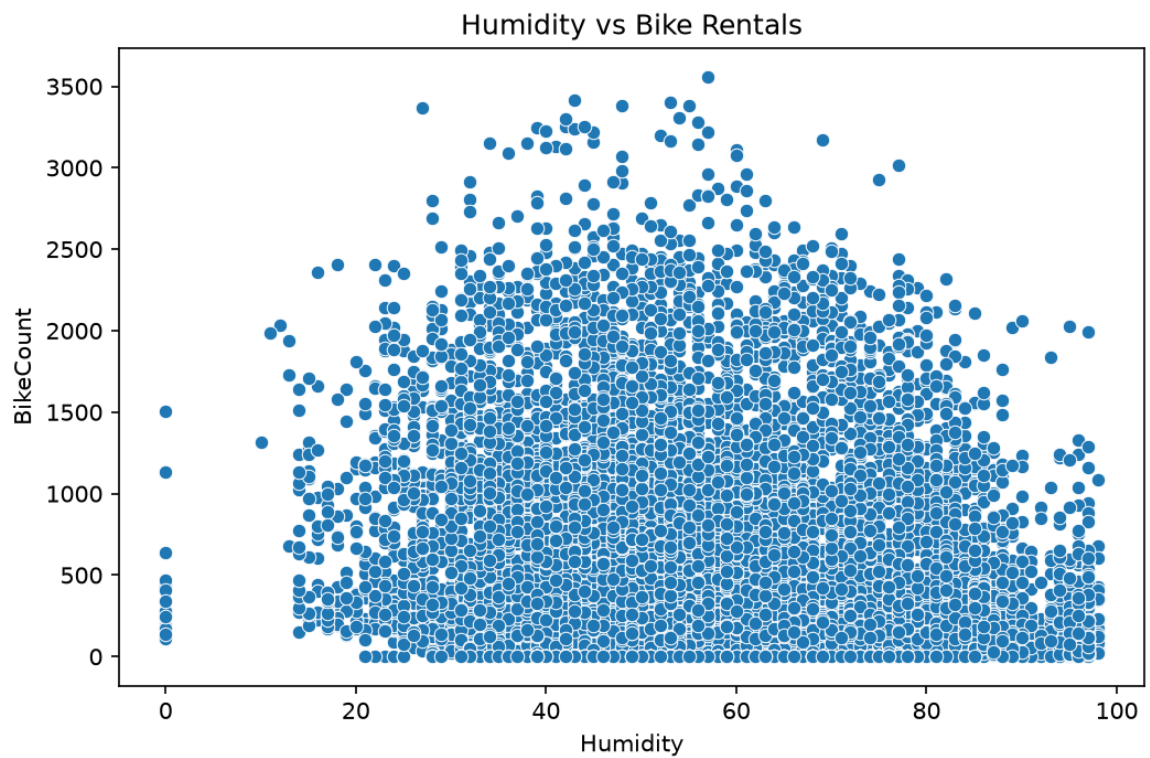


4. Humidity vs Bike Rentals

A scatter plot was used to examine the influence of humidity on bike rentals.

Observations

- Higher humidity levels are generally associated with lower bike rental demand.
- Demand decreases during highly humid weather conditions.
- The scatter plot indicates a negative relationship between humidity and bike usage.



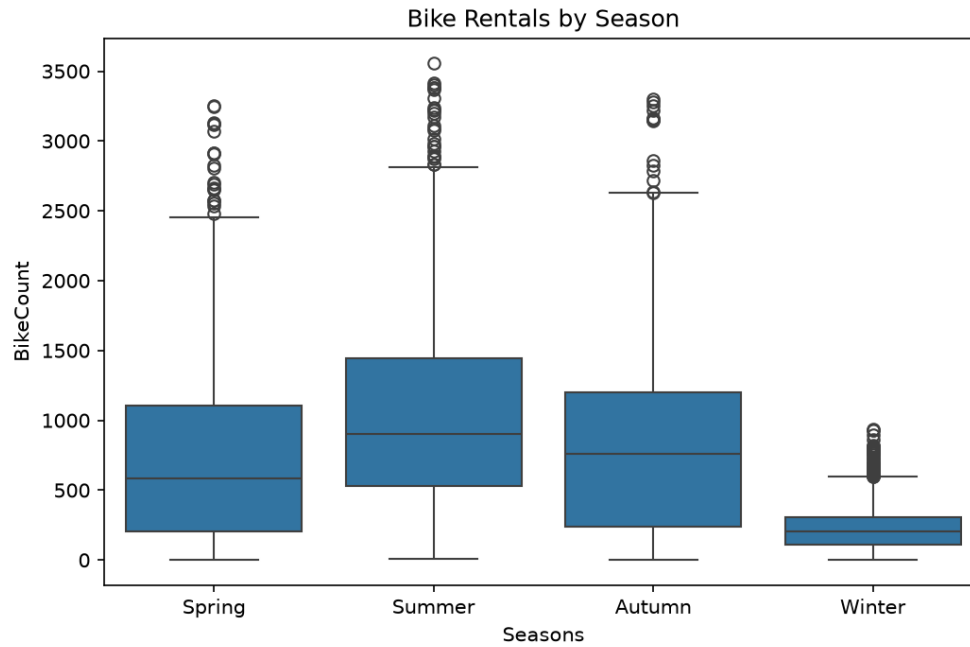
5. Bike Rentals by Season

A boxplot was generated to compare bike rental distributions across different seasons.

Observations

- Summer shows the highest rental activity.
- Autumn also experiences relatively high bike demand.

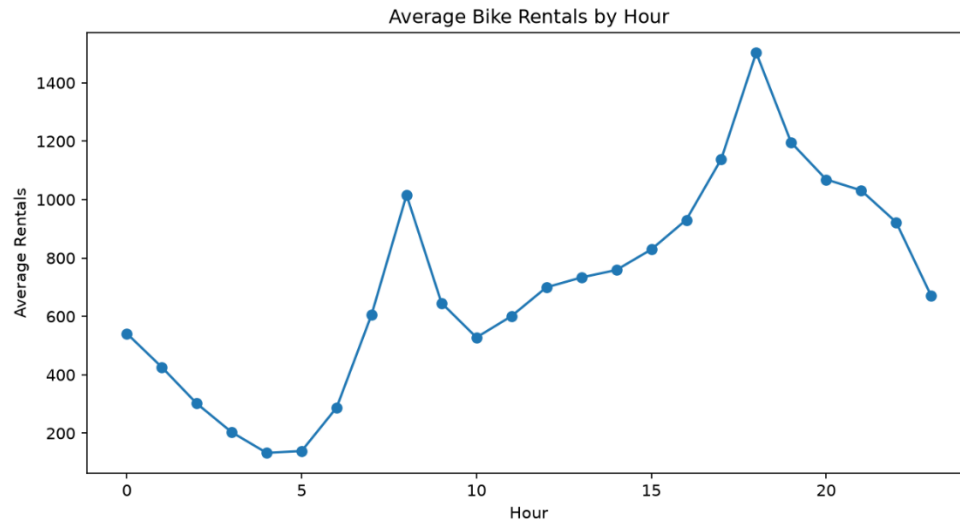
- Spring records moderate rental levels.
- Winter has the lowest bike rental demand because of colder weather.



6. Average Bike Rentals by Hour

The average number of bike rentals for each hour of the day was calculated and displayed using a line chart. **Observations**

- Rental demand increases during morning commuting hours.
- Demand reaches its highest levels during evening commuting hours.
- Late-night and early-morning hours show the lowest rental activity.
- Time of day is one of the most significant factors affecting bike rentals.



5.1.2 Machine Learning Model Performance

After completing data preprocessing and exploratory analysis, a **Linear Regression** model was developed to predict hourly bike rental demand.

Before model training, categorical variables (**Seasons**, **Holiday**, **FunctioningDay**, and **DayOfWeek**) were transformed using **One-Hot Encoding**, while numerical variables were passed directly to the model using a **ColumnTransformer**. A Scikit-learn **Pipeline** combined preprocessing and model training into a single workflow.

The dataset was divided into:

- **80% Training Data**
- **20% Testing Data**

The trained model generated predictions for the testing dataset. Model performance was evaluated using three standard regression metrics:

- **Mean Absolute Error (MAE)** measures the average difference between actual and predicted values.
- **Root Mean Squared Error (RMSE)** measures prediction error while assigning greater weight to larger errors.
- **R² Score** indicates how well the Linear Regression model explains the variation in bike rental demand.
- The performance of the model was evaluated using three standard regression metrics:

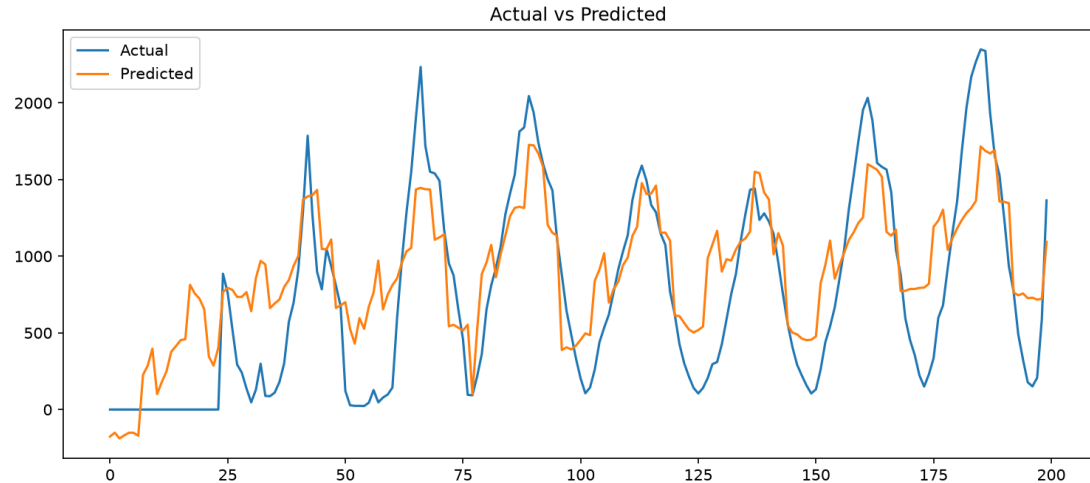
Evaluation Metric	Value
Mean Absolute Error (MAE)	277.05
Root Mean Squared Error (RMSE)	371.54
R ² Score	0.6330

The evaluation results obtained from the model indicate that the Linear Regression algorithm is capable of capturing the overall relationship between weather conditions, time-based variables, and bike rental demand.

Actual vs Predicted Bike Rentals

To evaluate prediction accuracy, the actual bike rental values were compared with the predicted values generated by the Linear Regression model. **Observations**

- Predicted values generally follow the overall trend of the actual bike rental demand.
- Minor differences exist because Linear Regression assumes a linear relationship between variables.
- The graph demonstrates that the model successfully captures the general demand pattern.



Residual Plot

A residual plot was generated to analyze prediction errors.

Observations

- Most residuals are scattered around the zero line.
- No strong systematic pattern is visible.
- The model does not exhibit significant prediction bias.

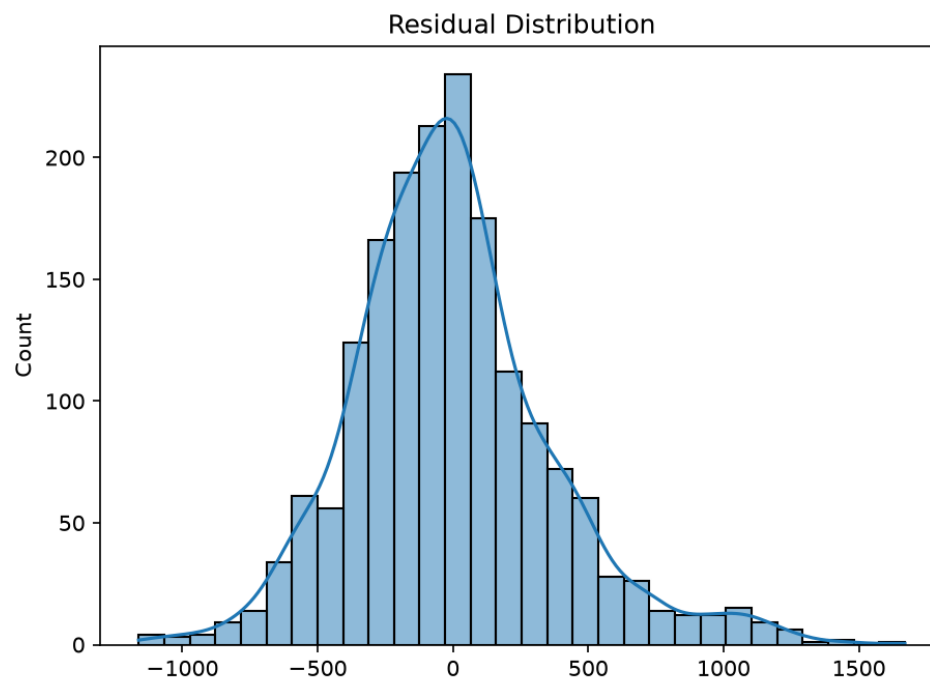


Residual Distribution

The distribution of residuals was visualized using a histogram.

Observations

- Residuals are approximately centered around zero.
- Most prediction errors are relatively small.
- The distribution indicates that the model performs reasonably well for overall demand prediction.



5.1.3 Power BI Dashboard Overview

After completing the Python implementation, the processed dataset and generated predictions were imported into Microsoft Power BI to create an interactive dashboard.

The dashboard enables users to analyze bike rental demand using various interactive visualizations and filters.

The dashboard includes:

- **Demand Overview**

Displays key performance indicators (KPIs) such as:

Total Bike Rentals

Average Bike Rentals

Peak Rental Hours

Seasonal Demand

- **Hourly Demand Analysis**

Shows hourly rental trends using line and area charts to identify periods of high and low demand.

- **Seasonal Analysis**

Compares bike rental demand across different seasons.

- **Weather Analysis**

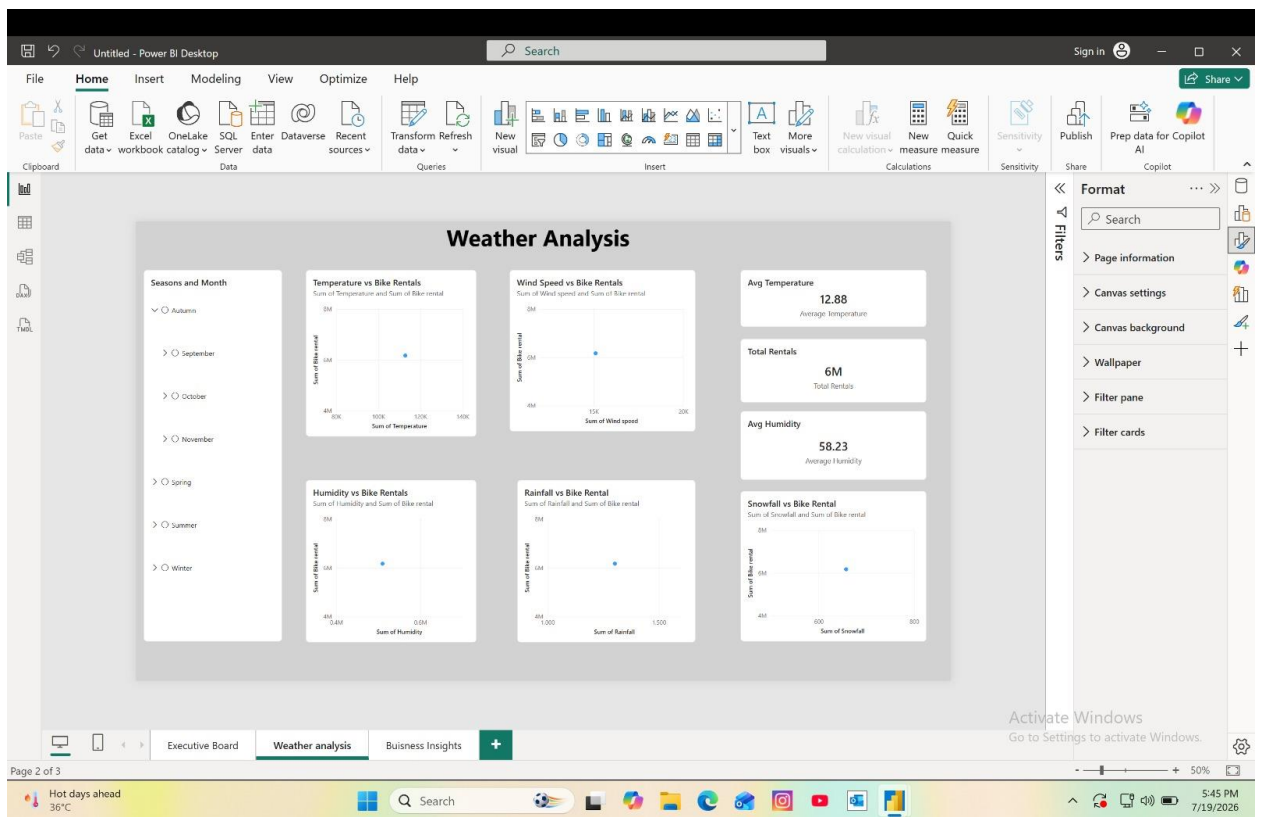
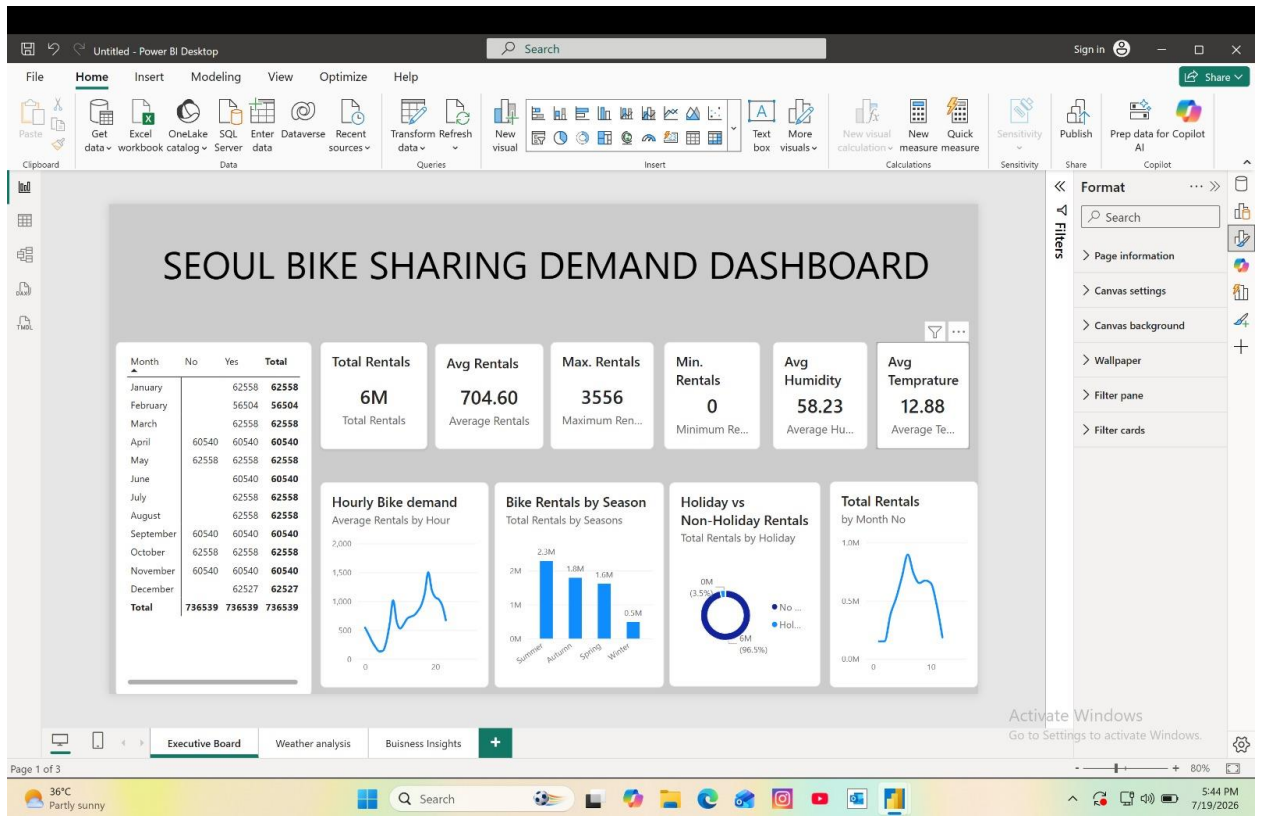
Illustrates the relationship between weather conditions such as temperature and humidity with bike rentals.

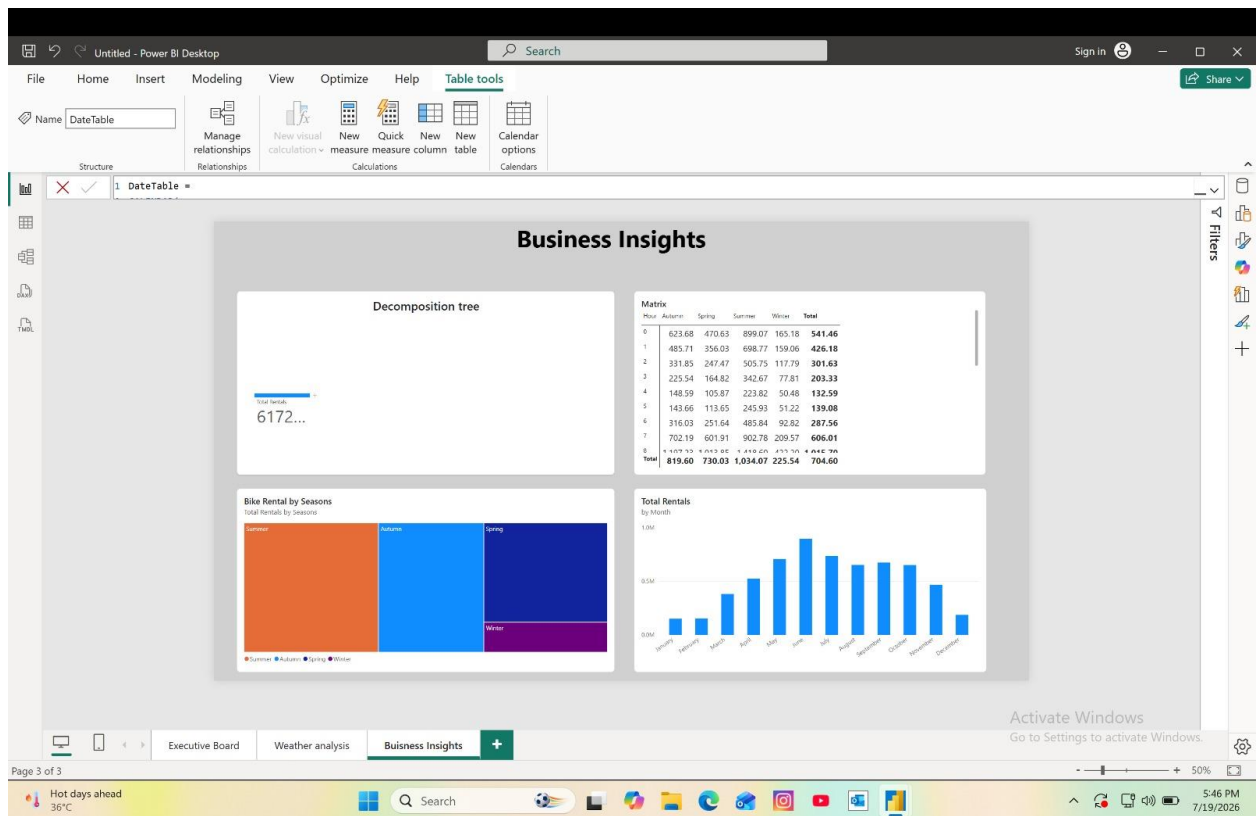
- **Interactive Filters**

Users can filter the dashboard using:

- Season
- Hour
- Holiday
- Functioning Day
- Day of Week

These interactive visualizations make it easier for users to explore rental patterns and support better operational decision-making.





5.2 Challenges Faced

During the implementation of the project, several challenges were encountered:

- Understanding the structure of the Seoul Bike Sharing dataset.
- Converting the Date column into datetime format for feature extraction.
- Selecting appropriate predictor variables for the machine learning model.
- Applying One-Hot Encoding to categorical variables.
- Building an effective preprocessing pipeline using Scikit-learn.
- Evaluating the prediction model using regression metrics.
- Designing meaningful visualizations in Microsoft Power BI.
- Interpreting the analytical results for business insights.

5.3 Key Learnings

The project provided valuable practical experience in data analytics and machine learning.

The major learnings include:

- Data cleaning and preprocessing using Pandas.

- Feature engineering using date and time information.
- Creating statistical visualizations with Matplotlib and Seaborn.
- Understanding relationships between weather variables and bike demand.
- Developing a Linear Regression model using Scikit-learn.
- Using a Pipeline and ColumnTransformer for efficient preprocessing.
- Evaluating regression models using MAE, RMSE, and R^2 Score.
- Creating interactive dashboards using Microsoft Power BI.
- Presenting analytical findings in a structured and business-oriented manner.

5.4 Entire Code Snippet

The complete Python implementation used to produce the results in this chapter is reproduced below.

```
C:\Users\DELL > Downloads > Project_Final.py > ...
4   import seaborn as sns
5
6   from sklearn.compose import ColumnTransformer
7   from sklearn.preprocessing import OneHotEncoder
8   from sklearn.pipeline import Pipeline
9   from sklearn.linear_model import LinearRegression
10  from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
11
12  df = pd.read_csv("SeoulBikeData.csv", encoding="latin1")
13
14  print(df.head())
15  print(df.describe())
16  print("Missing values:\n", df.isnull().sum())
17  print("Duplicate rows:", df.duplicated().sum())
18
19  df["Date"] = pd.to_datetime(df["Date"], format="%d/%m/%Y", errors="coerce")
20  df = df.drop_duplicates()
21  df = df.dropna(subset=["Date"])
22
23  df.rename(columns={
24      "Temperature(°C)": "Temperature",
25      "Humidity(%)": "Humidity",
26      "Wind speed (m/s)": "WindSpeed",
27      "Visibility (10m)": "Visibility",
28      "Dew point temperature(°C)": "DewPoint",
29      "Solar Radiation (MJ/m2)": "SolarRadiation",
30      "Rainfall(mm)": "Rainfall",
31      "Snowfall (cm)": "Snowfall",
32      "Functioning Day": "FunctioningDay",
33      "Rented Bike Count": "BikeCount"
34  }, inplace=True)
35
36  df["Month"] = df["Date"].dt.month_name()
37  df["DayOfWeek"] = df["Date"].dt.day_name()
38  df["Weekend"] = df["Date"].dt.dayofweek.apply(lambda x: 1 if x >= 5 else 0)
39  df["PeakHour"] = df["Hour"].apply(lambda x: 1 if (7 <= x <= 9 or 17 <= x <= 20) else 0)
40
41  plt.figure(figsize=(8, 5))
42  sns.histplot(df["BikeCount"], bins=30, kde=True)
43  plt.title("Distribution of Bike Rentals")
44  plt.show()
45
46  plt.figure(figsize=(10, 7))
47  sns.heatmap(df.select_dtypes(include=np.number).corr(), annot=True, cmap="coolwarm", fmt=".2f")
48  plt.title("Correlation Heatmap")
49  plt.show()
50
51  plt.figure(figsize=(8, 5))
52  sns.scatterplot(data=df, x="Temperature", y="BikeCount")
53  plt.title("Temperature vs Bike Rentals")
54  plt.show()
55
56  plt.figure(figsize=(8, 5))
57  sns.scatterplot(data=df, x="Humidity", y="BikeCount")
58  plt.title("Humidity vs Bike Rentals")
59  plt.show()
60
61  plt.figure(figsize=(8, 5))
62  sns.boxplot(data=df, x="Seasons", y="BikeCount",
63             order=["Spring", "Summer", "Autumn", "Winter"])
64  plt.title("Bike Rentals by Season")
65  plt.show()
```

```

plt.figure(figsize=(8, 5))
sns.boxplot(data=df, x="Seasons", y="BikeCount",
            order=["Spring", "Summer", "Autumn", "Winter"])
plt.title("Bike Rentals by Season")
plt.show()

plt.figure(figsize=(10, 5))
avg_hour = df.groupby("Hour")["BikeCount"].mean()
plt.plot(avg_hour.index, avg_hour.values, marker="o")
plt.title("Average Bike Rentals by Hour")
plt.xlabel("Hour")
plt.ylabel("Average Rentals")
plt.show()

features = [
    "Hour",
    "Temperature",
    "Humidity",
    "WindSpeed",
    "Visibility",
    "DewPoint",
    "SolarRadiation",
    "Rainfall",
    "Snowfall",
    "Seasons",
    "Holiday",
    "FunctioningDay",
    "DayOfWeek",
    "Weekend",
    "PeakHour"
]

```

```

X = df[features]
y = df["BikeCount"]

cat_cols = ["Seasons", "Holiday", "FunctioningDay", "DayOfWeek"]
num_cols = [c for c in features if c not in cat_cols]

split = int(len(df) * 0.8)

X_train = X.iloc[:split]
X_test = X.iloc[split:]
y_train = y.iloc[:split]
y_test = y.iloc[split:]

preprocessor = ColumnTransformer([
    ("cat", OneHotEncoder(handle_unknown="ignore"), cat_cols),
    ("num", "passthrough", num_cols)
])

model = Pipeline([
    ("preprocessor", preprocessor),
    ("regressor", LinearRegression())
])

model.fit(X_train, y_train)

pred = model.predict(X_test)

mae = mean_absolute_error(y_test, pred)
rmse = np.sqrt(mean_squared_error(y_test, pred))
r2 = r2_score(y_test, pred)

```

```

print("\nModel Evaluation")
print("MAE :", mae)
print("RMSE:", rmse)
print("R2  :", r2)

plt.figure(figsize=(12, 5))
plt.plot(y_test.values[:200], label="Actual")
plt.plot(pred[:200], label="Predicted")
plt.title("Actual vs Predicted")
plt.legend()
plt.show()

residuals = y_test.values - pred

plt.figure(figsize=(7, 5))
plt.scatter(pred, residuals)
plt.axhline(0, color="red")
plt.xlabel("Predicted")
plt.ylabel("Residual")
plt.title("Residual Plot")
plt.show()

plt.figure(figsize=(7, 5))
sns.histplot(residuals, bins=30, kde=True)
plt.title("Residual Distribution")
plt.show()

prediction_df = pd.DataFrame({
    "Actual": y_test.values,
    "Predicted": pred
})

prediction_df.to_csv("bike_predictions.csv", index=False)

print("\nPredictions saved to bike_predictions.csv")
print("Project completed.")

```

Chapter 6: Conclusion and Future Scope

6.1 Summary

This project successfully executed a complete data science lifecycle for bike-sharing demand forecasting. The major steps and contributions are summarised below:

The project began with importing the Seoul Bike Sharing Demand dataset into Python. Data preprocessing was performed by converting the **Date** column into datetime format, removing duplicate records, handling invalid dates, and renaming columns for better readability. Feature engineering was then carried out by creating new variables such as **Month**, **DayOfWeek**, **Weekend**, and **PeakHour**, which helped capture important temporal patterns in bike rental demand.

Exploratory Data Analysis (EDA) was conducted using various visualizations, including histograms, correlation heatmaps, scatter plots, boxplots, and hourly trend charts. These analyses revealed that weather conditions and time-based variables significantly influence bike rental demand.

A **Linear Regression** model was developed using a Scikit-learn Pipeline. Categorical variables were encoded using One-Hot Encoding, while numerical variables were processed directly. The model was evaluated using **Mean Absolute Error (MAE)**, **Root Mean Squared Error (RMSE)**, and **R² Score**, achieving the following results:

- **MAE: 277.05**
- **RMSE: 371.54**
- **R² Score: 0.6330**

These results indicate that the model successfully captures the general relationship between weather conditions, seasonal factors, time-based variables, and bike rental demand, explaining approximately **63.3% of the variation** in hourly rentals.

Finally, the processed data and prediction results were visualized using Microsoft Power BI to create an interactive dashboard for analyzing bike rental demand.

6.2 Conclusion

This project highlights the transformative role of predictive analytics and visualization in urban mobility operations:

The combination of data preprocessing, feature engineering, exploratory data analysis, and machine learning enabled the development of a predictive model capable of estimating hourly bike rental demand. The **Linear Regression** model achieved a **Mean Absolute Error (MAE) of 277.05**, a **Root Mean Squared Error (RMSE) of 371.54**, and an **R² Score of 0.6330**, indicating that it explains approximately **63.3%** of the variation in bike rental demand. These results show that the model provides a reliable baseline for forecasting hourly bike rentals using weather and temporal features.

The Power BI dashboard further enhanced the usefulness of the project by presenting analytical insights through interactive visualizations. The dashboard enables users to explore rental trends, identify peak demand periods, and understand the influence of weather conditions and seasonal variations.

Overall, the project provided valuable practical experience in data analytics, machine learning, and business intelligence while demonstrating how predictive analytics can support better planning and resource management in bike-sharing systems.

6.3 Future Scope

- Although the developed system provides useful insights and demand predictions, several improvements can further enhance its performance and usability.
- Future enhancements may include:
- Implementing advanced regression algorithms such as **Decision Tree Regression**, **Random Forest Regression**, or **Gradient Boosting Regression** to improve prediction accuracy and compare their performance with Linear Regression.
- Incorporating additional variables such as public events, traffic conditions, fuel prices, and weather forecasts.
- Developing a real-time demand forecasting system using live weather and rental data.
- Deploying the prediction model as a web application using **Flask** or **Streamlit**.
- Integrating the system with cloud platforms for scalable deployment and storage.
- Automating Power BI dashboards by connecting them directly to live datasets.
- Expanding the project to support demand forecasting for bike-sharing systems in multiple cities.
- Real-Time API: Deploy the best-performing model as a REST API (e.g., using Flask or FastAPI) to integrate live demand forecasts with fleet-management systems.

Bibliography & References

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- Scikit-learn: Machine Learning in Python.
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- Microsoft Power BI documentation.